

# Machine Learning-Based Buyer Segmentation and Investment Profiling for Real Estate Market Intelligence

Research Paper / Project Report

**Prepared for:** Unified Mentor Internship Project

**Domain:** Financial Analytics & Real Estate Market Intelligence

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**Project Dataset:** 2,000 real-estate buyers

**Machine Learning:** K-Means Clustering

This report documents the completed analytical workflow, cluster profiles, business insights, and Streamlit deployment for the real-estate buyer segmentation project.

GitHub: [github.com/Praveshyadav47/real-estate-buyer-segmentation](https://github.com/Praveshyadav47/real-estate-buyer-segmentation)

Live Dashboard: [pravesh-real-estate-segmentation.streamlit.app](https://pravesh-real-estate-segmentation.streamlit.app)

# Abstract

Real-estate markets contain buyers with different demographic characteristics, investment behaviors, property ownership levels, financing patterns, acquisition channels, and satisfaction levels. Treating these buyers as a single population can make marketing and investment analysis less targeted. This project applies unsupervised machine learning to identify meaningful buyer segments and convert the resulting clusters into business-oriented profiles.

Using a dataset of 2,000 buyers, the completed analysis produced three K-Means clusters. Cluster 0 contains 1,083 buyers (54.15%), Cluster 1 contains 867 buyers (43.35%), and Cluster 2 contains 50 buyers (2.50%). Cluster 2 has the highest average investment and property ownership, while Cluster 1 shows the highest average property value and property size. The project also examines geography, acquisition purpose, loan usage, referral channels, client type, and gender across the clusters.

The results are presented through an interactive Streamlit dashboard and supported by a GitHub repository containing the project notebook, application code, dashboard data, requirements, and documentation.

Keywords: Real Estate Analytics, Buyer Segmentation, K-Means, Clustering, Investment Profiling, Python, Streamlit

## 1. Introduction

The project requirements document identifies a need for data-driven understanding of different property buyer types, investment motivations, geographic differences, and customer financing patterns. It describes the business consequences of generic targeting as inefficient marketing spend, generic property recommendations, and poor investor targeting.

The project therefore uses clustering to discover groups of buyers with similar characteristics. The objective is not to predict a predefined label, but to explore hidden structure in the buyer population and translate that structure into actionable market intelligence.

## 2. Problem Statement

Real-estate businesses need to understand how buyers differ in their investment behavior, property ownership, demographics, geographic distribution, financing, acquisition purpose, and satisfaction. Without segmentation, marketing and recommendation strategies may treat materially different buyer groups in the same way.

## 3. Objectives

- Prepare and analyze buyer-level real-estate data.
- Identify meaningful buyer groups using unsupervised machine learning.
- Profile the resulting clusters using investment, property, demographic, and satisfaction measures.
- Study geographic and acquisition-channel patterns across clusters.
- Translate cluster-level findings into business recommendations.
- Build and deploy an interactive Streamlit dashboard for communicating the results.

## 4. Dataset and Variables

The project requirements document describes the source data using fields such as client identifier, client type, gender, country, region, date of birth/age indicator, acquisition purpose, loan applied, referral channel, and satisfaction score. These variables support demographic, geographic, financing, acquisition, and customer-experience analysis.

| Field / Concept     | Description                    |
|---------------------|--------------------------------|
| client_id           | Unique client identifier       |
| client_type         | Individual / Corporate         |
| gender              | Gender of buyer                |
| country             | Country of residence           |
| region              | Geographic region              |
| date_of_birth / age | Age indicator                  |
| acquisition_purpose | Investment / Personal use      |
| loan_applied        | Financing indicator            |
| referral_channel    | Source of customer acquisition |
| satisfaction_score  | Customer satisfaction rating   |

## 5. Data Preparation and Methodology

The project workflow followed the methodology described in the requirements document: data cleaning, feature preparation/encoding, feature scaling, clustering model selection, cluster evaluation, and interpretation. The PRD specifically identifies one-hot/label encoding for categorical variables and StandardScaler or MinMaxScaler for numeric variables.

### 5.1 Data Cleaning

- Handle missing client attributes where required.
- Normalize categorical labels.
- Remove duplicate client entries where identified.

### 5.2 Feature Preparation and Scaling

- Prepare categorical variables for analysis using appropriate encoding.
- Scale numeric variables so that variables with larger numeric ranges do not dominate distance-based clustering.
- The deployed dashboard uses six core analysis features: age, total properties, total investment, average property value, average property size, and satisfaction score.

### 5.3 Clustering

K-Means clustering was used as the primary segmentation method. K-Means assigns observations to groups by minimizing within-cluster variation around cluster centroids. The project requirements document also identifies hierarchical clustering as a possible validation approach.

### 5.4 Cluster Selection and Interpretation

The requirements document recommends the Elbow Method and Silhouette Score for evaluating the number and quality of clusters. After clustering, each group is interpreted using investment purpose, geography, loan behavior, demographics, property characteristics, and satisfaction.

## 6. Results and Cluster Profiles

The completed analysis contains 2,000 buyers divided into three clusters. The cluster labels are data-derived identifiers; they should not be confused with the four illustrative buyer types listed in the project requirements document. The actual completed model used three clusters.

| Metric                 | Cluster 0 | Cluster 1 | Cluster 2 |
|------------------------|-----------|-----------|-----------|
| Buyer count            | 1,083     | 867       | 50        |
| Share                  | 54.15%    | 43.35%    | 2.50%     |
| Average age            | 55.55     | 54.88     | 68.05     |
| Average properties     | 3.54      | 3.59      | 7.32      |
| Average investment     | 1,050,597 | 1,455,737 | 2,416,603 |
| Average property value | 298,060   | 409,162   | 332,747   |
| Average property size  | 992.03    | 1,344.33  | 1,101.05  |
| Average satisfaction   | 2.96      | 3.09      | 3.52      |

### 6.1 Cluster 0 — Largest General Buyer Segment

Cluster 0 is the largest segment, containing 1,083 buyers (54.15%). Its average investment is approximately 1.05 million, with an average of 3.54 properties per buyer. The average satisfaction score is 2.96, the lowest among the three clusters. This makes satisfaction-related analysis a useful area for further investigation.

### 6.2 Cluster 1 — Higher-Value Property Profile

Cluster 1 contains 867 buyers (43.35%). It has an average investment of approximately 1.46 million, the highest average property value at approximately 409,162, and the highest average property size at approximately 1,344.33. These characteristics support targeted analysis of higher-value and larger property preferences.

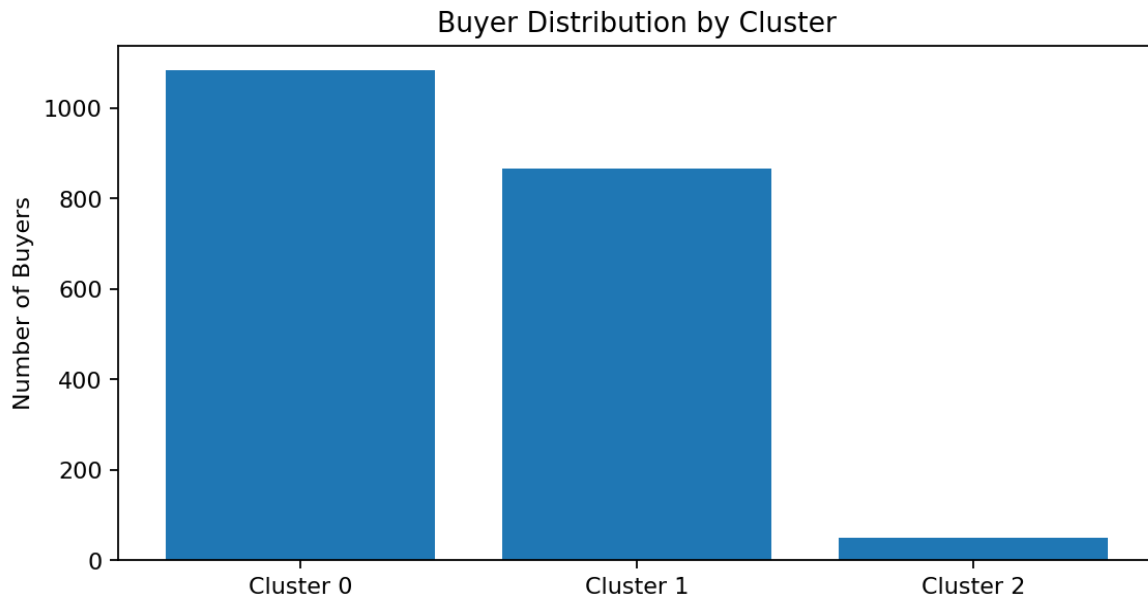
### 6.3 Cluster 2 — Small, High-Investment Segment

Cluster 2 contains only 50 buyers (2.50%), but it has the highest average investment at approximately 2.42 million and the highest average property ownership at 7.32 properties. Its average satisfaction score is also the highest at 3.52. Because the segment is small but materially different in investment and ownership, it is useful to monitor it separately.

## 7. Visual Analysis

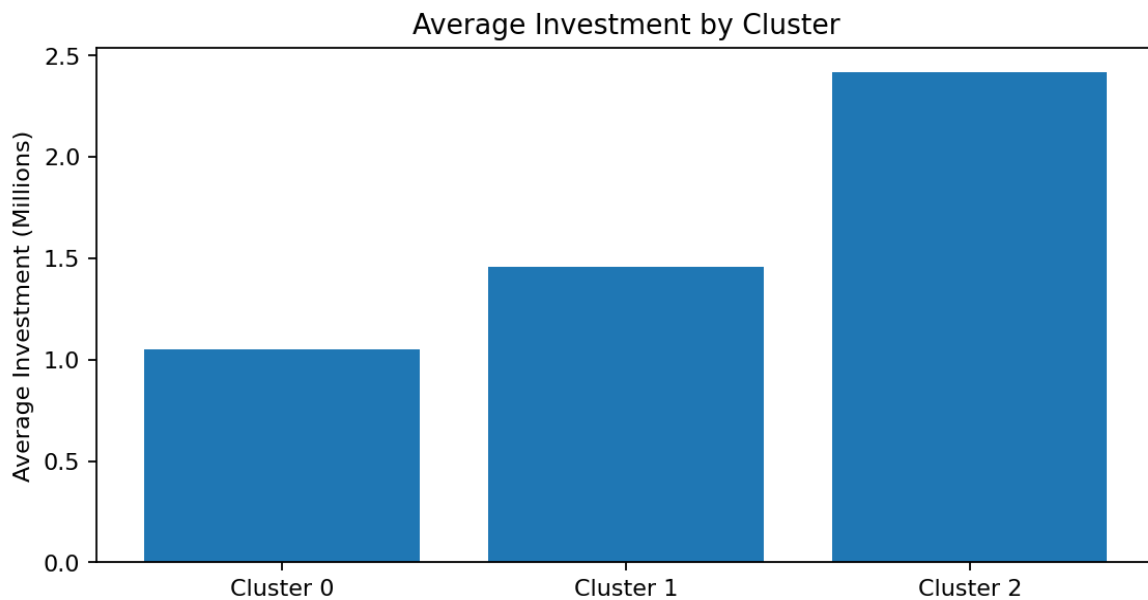
### 7.1 Buyer Distribution

The cluster distribution is highly concentrated in Clusters 0 and 1, while Cluster 2 represents a small specialist segment.

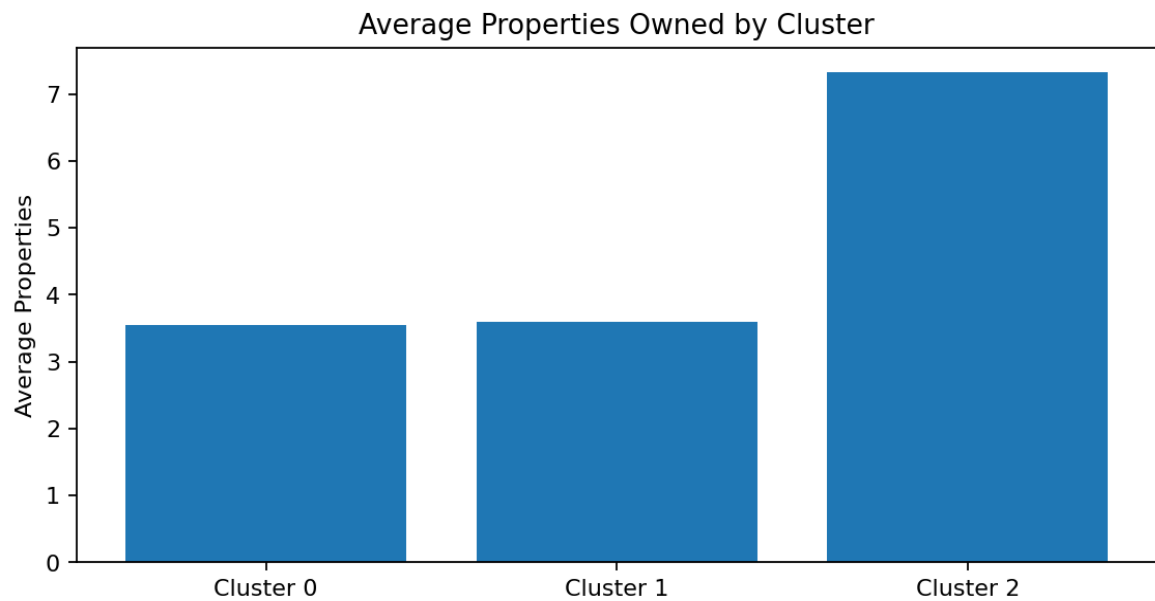


### 7.2 Investment and Property Ownership

Average investment rises substantially from Cluster 0 to Cluster 2. Cluster 2 also has more than twice the average property ownership of Cluster 0.

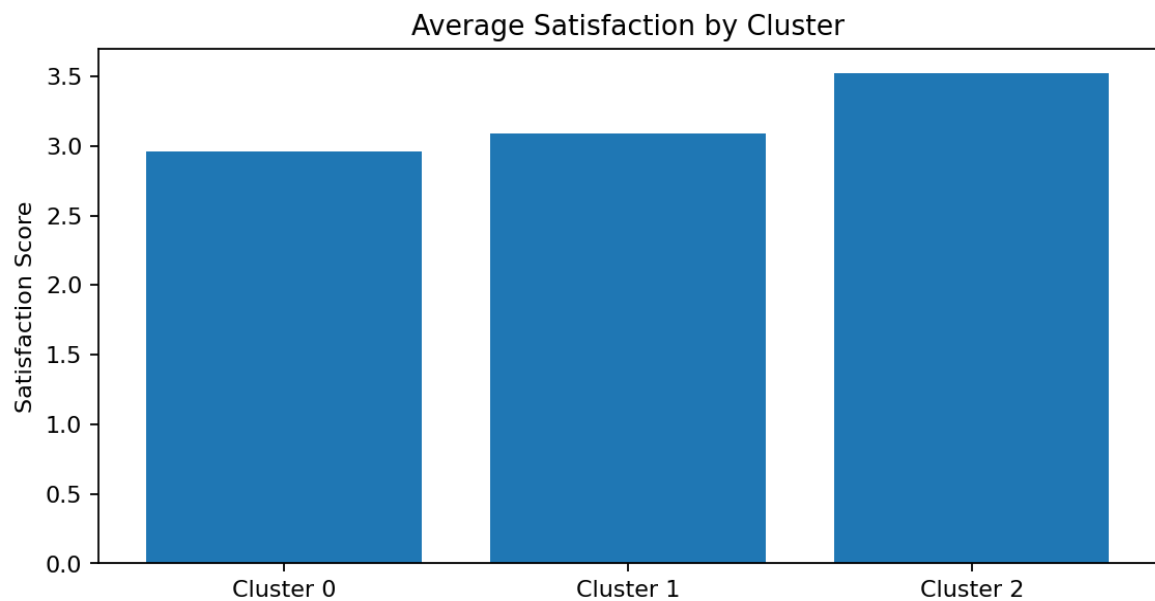


## 7.3 Property Ownership



## 7.4 Satisfaction

Cluster 2 has the highest average satisfaction, while Cluster 0 has the lowest average satisfaction.



## 8. Geographic and Behavioral Analysis

### 8.1 Country Distribution

The USA is the dominant country across all three clusters. Its share is approximately 75.8% in Cluster 0, 77.7% in Cluster 1, and 86% in Cluster 2. Other countries represented in the data include Australia, Belgium, Canada, Denmark, France, Germany, Mexico, Russia, and the UK.

### 8.2 Regional Distribution

California appears as a prominent region across the clusters, including approximately 48% of Cluster 2. This indicates that geographic analysis can be useful when developing regional marketing or property strategies.

### 8.3 Acquisition Purpose

The clusters are broadly similar in acquisition purpose. Home/personal-use purchases account for about 69% in Cluster 0, 70% in Cluster 1, and 66% in Cluster 2. Investment purchases account for the remainder, with Cluster 2 showing the highest investment-purpose share at approximately 34%.

### 8.4 Loan Applications

Loan application rates are relatively similar across the clusters: approximately 37% in Cluster 0, 36% in Cluster 1, and 32% in Cluster 2. Cluster 2 therefore has a somewhat lower observed loan-application share alongside its higher average investment.

### 8.5 Referral Channel

Website referrals are the largest acquisition channel across all clusters. They account for approximately 54% of Cluster 0, 55% of Cluster 1, and 70% of Cluster 2. Agency referrals account for approximately 36%, 35%, and 26%, respectively, while client referrals are smaller.

### 8.6 Client Type and Gender

Individual buyers dominate all clusters, representing approximately 94–96% of each group. Gender shares are relatively balanced in Clusters 0 and 1, while Cluster 2 has a higher male share in the analyzed data.

## 9. Business Insights and Recommendations

### 1. Develop segment-specific marketing strategies

The three clusters differ materially in investment, property ownership, property value, property size, and satisfaction. Marketing and recommendation strategies can therefore be tailored to the characteristics of each group rather than treating all buyers identically.

### 2. Target higher-value property demand

Cluster 1 has the highest average property value and property size. Property campaigns for this segment can emphasize suitable higher-value and larger properties.

### 3. Use digital acquisition strategically

Website referrals are particularly prominent in Cluster 2. Improving digital property discovery, website content, and online conversion journeys may therefore be relevant when engaging high-investment buyers.

### 4. Monitor the high-investment segment separately

Cluster 2 is small but has the highest average investment and property ownership. It can be tracked as a specialized investment-oriented segment for further research and targeted relationship management.

### 5. Investigate satisfaction drivers

Cluster 0 has the lowest average satisfaction score. Further analysis of property experience, acquisition journey, service interactions, and other available variables could help explain this difference.

## 10. Streamlit Application

An interactive Streamlit dashboard was developed to communicate the segmentation results. The dashboard presents the project title and methodology, a dataset overview, total buyers, number of clusters, feature count, and buyer cluster distribution. The application uses the committed dashboard data files and is deployed through Streamlit Community Cloud.

The current live dashboard is available at: <https://pravesh-real-estate-segmentation.streamlit.app/>

Streamlit Community Cloud supports deploying an app by selecting a GitHub repository, branch, and entrypoint file, and provides a shareable streamlit.app URL. The project's deployed configuration uses the GitHub repository, the main branch, and `app/app.py` as the entrypoint. ■cite■turn0search0■

## 11. Limitations and Future Scope

- The clustering results describe patterns in the analyzed dataset and should not automatically be treated as causal relationships.
- Cluster labels such as Cluster 0, Cluster 1, and Cluster 2 are identifiers rather than inherent buyer categories.
- The smallest cluster contains only 50 buyers, so additional data would help assess whether this segment is stable across samples or time periods.
- Future versions can compare K-Means with hierarchical clustering as suggested in the project requirements.
- Future work can add richer dashboard filters and geographic visualizations using the country and region variables described in the requirements document.

## 12. Conclusion

This project demonstrates how unsupervised machine learning can be used to convert real-estate buyer data into actionable market segments. K-Means clustering identified three groups with distinct investment, property ownership, property value, property size, and satisfaction characteristics. Additional profiling revealed geographic and acquisition-channel patterns that can support more focused marketing and investor analysis.

The project also converts the analytical workflow into a usable Streamlit application and publishes the work through GitHub and Streamlit Community Cloud. This provides a complete data-analytics workflow from data preparation and machine learning through business interpretation and deployment.

## 13. Project Links

**GitHub Repository:** <https://github.com/Praveshyadav47/real-estate-buyer-segmentation>

**Live Streamlit App:** <https://pravesh-real-estate-segmentation.streamlit.app/>

## 14. References

1. Unified Mentor / Parcl Co. Limited — Project Requirements Document: *Machine learning based Buyer Segmentation and Investment Profiling for Real Estate Market Intelligence*.
2. Streamlit Documentation — Deploy your app on Community Cloud:  
<https://docs.streamlit.io/deploy/streamlit-community-cloud/deploy-your-app>
3. Project analysis and completed implementation: Jupyter Notebook, GitHub repository, and deployed Streamlit application.

End of Report